

EXPERIMENT - 17

1. Problem Statement

Write a menu-driven C program using `switch-case` to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, with void return type and with a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result.

- Generate the largest possible number using its digits.
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

2. Algorithm

Step 1: START

Step 2: Display the menu:

1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number

Step 3: Read the choice entered by the user.

Step 4: Using switch (choice):

- case 1: call `largest_number()` and break
- case 2: call `smallest_number()` and break
- case 3: call `swap_first_last()` and break
- case 4: call `check_palindrome()` and break
- default: print "Invalid choice" and break

Step 5: STOP

Function `largest_number()`

Step 1: Read num.

Step 2: If num is less than or equal to 0, print the error message and return.

Step 3: Make `count[0]` to `count[9]` equal to 0, and set `temp = num`.

Step 4: Repeat while temp is greater than 0:

```
digit = temp % 10
count[digit] = count[digit] + 1
temp = temp / 10
```

Step 5: Set `result = 0`.

Step 6: For i from 9 down to 0, repeat `count[i]` times: `result = result * 10 + i`.

Step 7: Print result as the largest number.

Function `smallest_number()`

Step 1: Read num.

Step 2: If num is less than or equal to 0, print the error message and return.

Step 3: Count the digits in `count[0]` to `count[9]` in the same way as above.

Step 4: Find the smallest non-zero digit, because the first digit of a number cannot be 0. For i from 1 to 9, if count[i] is greater than 0 then take first = i and stop the loop.

Step 5: Set result = first and reduce count[first] by 1.

Step 6: For i from 0 to 9, repeat count[i] times: result = result * 10 + i.

Step 7: Print result as the smallest number.

Function swap_first_last()

Step 1: Read num.

Step 2: If num is less than or equal to 0, print the error message and return.

Step 3: Count how many digits num has.

Step 4: If there is only one digit, then result = num.

Step 5: Otherwise:

divisor = 10 raised to the power (digits – 1)

first = num / divisor

last = num % 10

middle = (num % divisor) / 10

result = last * divisor + middle * 10 + first

Step 6: Print result as the number after swapping.

Function check_palindrome()

Step 1: Read num.

Step 2: If num is less than or equal to 0, print the error message and return.

Step 3: Set rev = 0 and temp = num.

Step 4: Repeat while temp is greater than 0:

digit = temp % 10

rev = rev * 10 + digit

temp = temp / 10

Step 5: If rev is equal to num, print "Palindrome number", otherwise print "Not a palindrome number".

3. Test Case Table

TC #	Choice	Number	Expected Output	Actual Output	Result
TC1	1	5382	Largest number = 8532	Largest number = 8532	PASS
TC2	1	1020	Largest number = 2100	Largest number = 2100	PASS
TC3	1	777	Largest number = 777	Largest number = 777	PASS
TC4	2	5382	Smallest number = 2358	Smallest number = 2358	PASS
TC5	2	1020	Smallest number = 1002	Smallest number = 1002	PASS
TC6	2	7	Smallest number = 7	Smallest number = 7	PASS
TC7	3	5382	After swapping = 2385	After swapping = 2385	PASS
TC8	3	12	After swapping = 21	After swapping = 21	PASS
TC9	3	7	After swapping = 7	After swapping = 7	PASS
TC10	3	1020	After swapping = 21	After swapping = 21	PASS
TC11	4	12321	Palindrome number	Palindrome number	PASS
TC12	4	1234	Not a palindrome number	Not a palindrome number	PASS
TC13	4	7	Palindrome number	Palindrome number	PASS
TC14	4	1221	Palindrome number	Palindrome number	PASS
TC15	1	-5	Error message	Error: -5 is not a positive integer.	PASS
TC16	4	0	Error message	Error: 0 is not a positive integer.	PASS
TC17	9	-	Invalid choice message	Invalid choice! Enter 1 to 4.	PASS

4. Source Code

File name: `number_menu.c` (202 lines)

```
/* Menu-driven program using switch-case to perform digit based
operations on a positive integer.
One separate user-defined function is written for every menu
option. All the functions have void return type and are called
by a simple function call, without any parameter. */

#include <stdio.h>

void largest_number(void);
void smallest_number(void);
void swap_first_last(void);
void check_palindrome(void);

int main(void)
{
    int choice;

    printf("===== MENU =====\n");
    printf("1. Generate the largest possible number using its digits\n");
    printf("2. Generate the smallest possible number using its digits\n");
    printf("3. Swap its first and last digits\n");
    printf("4. Check whether it is a palindrome number\n");
    printf("===== \n");
    printf("Enter your choice (1 to 4): ");
    scanf("%d", &choice);

    switch (choice) {

        case 1:
            largest_number();
            break;

        case 2:
            smallest_number();
            break;

        case 3:
            swap_first_last();
            break;

        case 4:
            check_palindrome();
            break;

        default:
            printf("\nInvalid choice! Enter 1 to 4.\n");
            break;
    }

    return 0;
}
```

```

/* Option 1 : largest possible number using the same digits */
void largest_number(void)
{
    long num, temp, result;
    int digit, i, j;
    int count[10];

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    for (i = 0; i < 10; i++)
        count[i] = 0;

    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        count[digit]++;
        temp = temp / 10;
    }

    result = 0;
    for (i = 9; i >= 0; i--) {
        for (j = 1; j <= count[i]; j++)
            result = result * 10 + i;
    }

    printf("\nNumber          : %ld\n", num);
    printf("Largest number : %ld\n", result);
}

/* Option 2 : smallest possible number using the same digits */
void smallest_number(void)
{
    long num, temp, result;
    int digit, i, j, first;
    int count[10];

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    for (i = 0; i < 10; i++)
        count[i] = 0;

    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        count[digit]++;
        temp = temp / 10;
    }

    /* the first digit cannot be 0, so the smallest non-zero

```

```

        digit is placed first and then the zeros are placed    */
first = 0;
for (i = 1; i <= 9; i++) {
    if (count[i] > 0) {
        first = i;
        break;
    }
}

result = first;
count[first]--;

for (i = 0; i <= 9; i++) {
    for (j = 1; j <= count[i]; j++)
        result = result * 10 + i;
}

printf("\nNumber          : %ld\n", num);
printf("Smallest number : %ld\n", result);
}

/* Option 3 : swap the first and the last digit */
void swap_first_last(void)
{
    long num, temp, divisor, middle, result;
    int digits, first, last, i;

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    digits = 0;
    temp = num;
    while (temp > 0) {
        digits++;
        temp = temp / 10;
    }

    if (digits == 1) {
        result = num;
    }
    else {
        divisor = 1;
        for (i = 1; i <= digits - 1; i++)
            divisor = divisor * 10;

        first = num / divisor;
        last = num % 10;
        middle = (num % divisor) / 10;

        result = last * divisor + middle * 10 + first;
    }

    printf("\nNumber          : %ld\n", num);
    printf("After swapping : %ld\n", result);
}

```

```
/* Option 4 : check whether the number is a palindrome */
void check_palindrome(void)
{
    long num, temp, rev;
    int digit;

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    rev = 0;
    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        rev = rev * 10 + digit;
        temp = temp / 10;
    }

    printf("\nNumber : %ld\n", num);
    if (rev == num)
        printf("Output : Palindrome number\n");
    else
        printf("Output : Not a palindrome number\n");
}
```

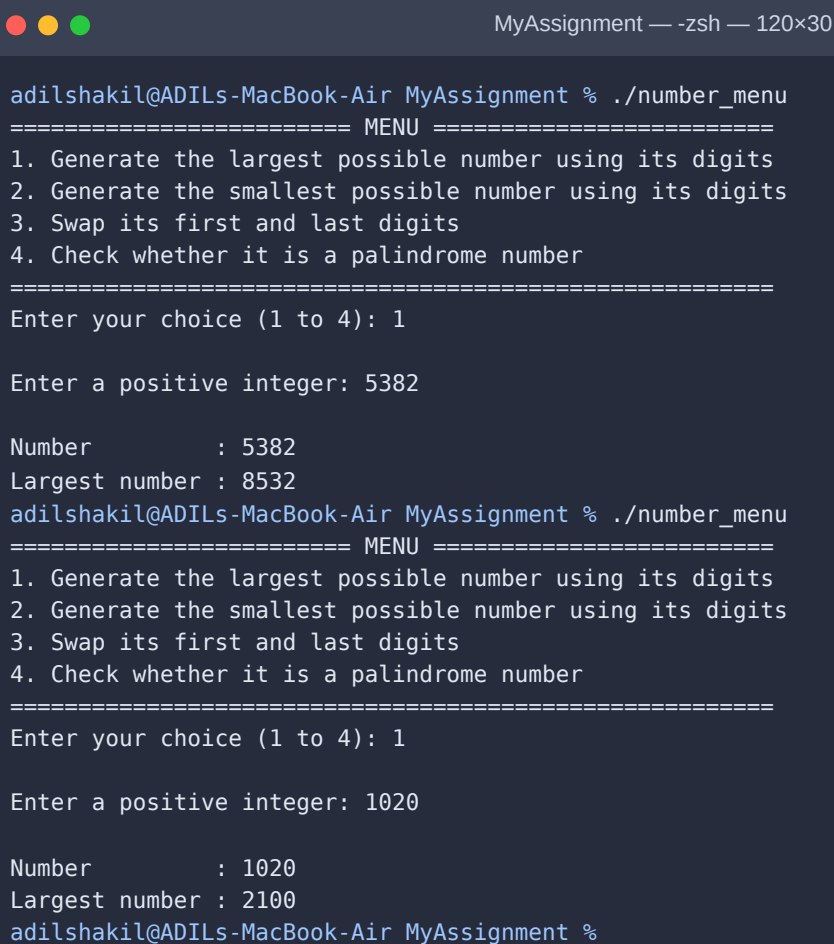
5. Compilation and Output

I wrote the program in the `MyAssignment` folder using the `nano` editor, then compiled and ran it:

```
$ cd ~/Desktop/MyAssignment
$ nano number_menu.c
$ gcc number_menu.c -o number_menu
$ ./number_menu
```

The program compiled in the first attempt itself, without any error or warning.

Option 1 : largest possible number



```
MyAssignment — zsh — 120x30

adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 1

Enter a positive integer: 5382

Number          : 5382
Largest number  : 8532
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 1

Enter a positive integer: 1020

Number          : 1020
Largest number  : 2100
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Option 2 : smallest possible number

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 2

Enter a positive integer: 5382

Number          : 5382
Smallest number : 2358
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 2

Enter a positive integer: 1020

Number          : 1020
Smallest number : 1002
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Option 3 : swap the first and last digit

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 3

Enter a positive integer: 5382

Number      : 5382
After swapping : 2385
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 3

Enter a positive integer: 12

Number      : 12
After swapping : 21
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Option 4 : palindrome check



```
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 4

Enter a positive integer: 12321

Number : 12321
Output : Palindrome number
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 4

Enter a positive integer: 1234

Number : 1234
Output : Not a palindrome number
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Invalid input and invalid choice



```
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 1

Enter a positive integer: -5

Error: -5 is not a positive integer.
adilshakil@ADILs-MacBook-Air MyAssignment % ./number_menu
===== MENU =====
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
=====
Enter your choice (1 to 4): 9

Invalid choice! Enter 1 to 4.
adilshakil@ADILs-MacBook-Air MyAssignment %
```